PROBLEM 23 | Pendulum & Liftoff

Simple Pendulum (Unilateral Constraint)

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A simple pendulum consists of a mass m hanging from a string of length L and fixed at a pivot point O. The simple pendulum is equivalent to the problem of a heavy particle constrained to move along a smooth circumference. A unilateral constraint is considered, such that the particle moves along the interior of the circumference and it might fall under certain conditions.

Analyze the motion of the particle when it is pushed from the lowest point of the circumference with velocity v_0. At what point does the particle leave the circumference? Is this point ever reached?

If the circumference rotates around the vertical axis with a constant angular velocity equal to ω, write down the new equations of motion. Compute the locations along the circumference in which the particle is in equilibrium.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle of mass m is suspended by a string of length L from pivot O, or equivalently moves along the smooth interior of a vertical circular track of radius L.
- **Unilateral Constraint:** The particle cannot cross the circular boundary outward, but may detach if the inward contact force drops to zero ($N = 0$, or string slack $T = 0$).
- **Initial Motion:** Pushed from the lowest point ($\theta = 0$) with velocity v_0.
- **Part (b) Modification:** The circular track rotates about its vertical diameter with constant angular velocity ω.
- **Parameters:** $m > 0$ [kg], $L > 0$ [m], $g > 0$ [m/s^2], $v_0 > 0$ [m/s], $\omega \geq 0$ [rad/s].

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial Frame $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$:** Oz points vertically upward ($-\mathbf{k}$ is downward).
- **Polar Coordinate Basis $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{j}\}$:**

Angle θ measured from the downward vertical ($-\mathbf{k}$):

$$\mathbf{e}_r = \sin\theta\,\mathbf{i} - \cos\theta\,\mathbf{k} \quad \text{(pointing radially outward)}$$

$$\mathbf{e}_\theta = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{k} \quad \text{(tangential, along increasing } \theta\text{)}$$

- **Kinematics:**
- Position: $\mathbf{r} = L\mathbf{e}_r$.
- Velocity: $\mathbf{v} = L\dot{\theta}\mathbf{e}_\theta$.
- Acceleration: $\mathbf{a} = -L\dot{\theta}^2\mathbf{e}_r + L\ddot{\theta}\mathbf{e}_\theta$.
- **Change of basis.** The triple $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{j}\}$ listed above has $\mathbf{e}_r \times \mathbf{e}_\theta = -\mathbf{j}$, so it is left-handed (determinant -1); the right-handed polar triple is $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_3\}$ with $\mathbf{e}_3 = -\mathbf{j}$ (the third vector plays no role in the planar kinematics). The columns are the components in $\mathcal{B}_0 = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$:

$$[{}_0R_1] = \begin{pmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \\ -\cos\theta & \sin\theta & 0 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- The angular velocity of $\mathcal{B}_1$ is $\boldsymbol{\omega}_{10} = \dot{\theta}\,\mathbf{e}_3 = -\dot{\theta}\,\mathbf{j}$, and Poisson's formula gives $\dot{\mathbf{e}}_r = \dot{\theta}\,\mathbf{e}_\theta$, $\dot{\mathbf{e}}_\theta = -\dot{\theta}\,\mathbf{e}_r$, the identities behind $\mathbf{v} = L\dot{\theta}\,\mathbf{e}_\theta$ and $\mathbf{a} = -L\dot{\theta}^2\mathbf{e}_r + L\ddot{\theta}\,\mathbf{e}_\theta$.
- For part (b) the track rotates about $\mathbf{k}$ at rate ω: the rotating basis $\{\mathbf{e}_\rho, \mathbf{e}_\phi, \mathbf{k}\}$ is related to $\mathcal{B}_0$ by $\mathrm{R}_z(\omega t)$ (a rotation matrix with the same properties, $\det = 1$, $RR^T = I$), and inside it the pair $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ is written with $\mathbf{e}_\rho$ in place of $\mathbf{i}$. The determinant and the product RR^T were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

Part (a): Planar Motion & Detachment Analysis

1. Mechanical Energy Conservation

Pedagogical Justification: The normal constraint force $\mathbf{N} = -N\mathbf{e}_r$ is always perpendicular to velocity $\mathbf{v} = L\dot{\theta}\mathbf{e}_\theta$ ($\mathbf{N} \cdot \mathbf{v} = 0$), so it does no work. Mechanical energy is strictly conserved.

Taking the lowest point $\theta = 0$ as the gravitational potential reference ($V(0) = 0$):

$$V(\theta) = mg(L - L\cos\theta) = mgL(1 - \cos\theta)$$

$$E = \frac{1}{2}mv_0^2 = \frac{1}{2}mv^2 + mgL(1 - \cos\theta)$$

Isolate $v^2(\theta)$:

$$v^2(\theta) = v_0^2 - 2gL(1 - \cos\theta)$$

2. Normal Reaction Force $N(\theta)$

Pedagogical Justification: Newton's second law along the inward radial normal $\mathbf{e}_n = -\mathbf{e}_r$ provides an explicit expression for the contact reaction force $N(\theta)$ as a function of angle θ.

The forces acting on the particle are gravity $\mathbf{F}_g = -mg\mathbf{k}$ and track reaction $\mathbf{N} = -N\mathbf{e}_r$ ($N > 0$ pushes inward):

$$\mathbf{F}_g = -mg\mathbf{k} = -mg(-\cos\theta\,\mathbf{e}_r + \sin\theta\,\mathbf{e}_\theta) = mg\cos\theta\,\mathbf{e}_r - mg\sin\theta\,\mathbf{e}_\theta$$

Projecting Newton's second law along the inward normal $-\mathbf{e}_r$:

$$\sum F_n = N - mg\cos\theta = ma_n = m\frac{v^2}{L}$$

$$N(\theta) = mg\cos\theta + \frac{m}{L}v^2(\theta)$$

Substitute the energy relation for $v^2(\theta)$:

$$N(\theta) = mg\cos\theta + \frac{m}{L}\left[v_0^2 - 2gL(1-\cos\theta)\right]$$

Expand and combine terms:

$$N(\theta) = mg\cos\theta + \frac{mv_0^2}{L} - 2mg + 2mg\cos\theta = \frac{mv_0^2}{L} - 2mg + 3mg\cos\theta$$

$$N(\theta) = \frac{mv_0^2}{L} - mg(2 - 3\cos\theta)$$

3. Detachment Condition ($N = 0$)

The particle leaves the circumference if N drops to zero while $v > 0$:

$$N(\theta_d) = 0 \implies \frac{mv_0^2}{L} - 2mg + 3mg\cos\theta_d = 0$$

$$3mg\cos\theta_d = 2mg - \frac{mv_0^2}{L} \implies \cos\theta_d = \frac{2gL - v_0^2}{3gL}$$

4. Motion Regimes and Question Resolution: "Is this point ever reached?"

- **Regime 1:** $v_0 \leq \quad 2gL$ **(Low energy, oscillation in lower semicircle)**

Velocity reaches zero when $v^2 = 0$:

$$v_0^2 - 2gL(1 - \cos\theta_{max}) = 0 \implies \cos\theta_{max} = 1 - \frac{v_0^2}{2gL} \geq 0 \implies \theta_{max} \leq \frac{\pi}{2}$$

At this maximum angle, the reaction force is:

$$N(\theta_{max}) = mg\cos\theta_{max} \geq 0$$

The particle comes to rest and reverses direction before N can vanish.

Conclusion: The detachment point is **never reached**. The particle oscillates periodically in the lower semicircle.

- **Regime 2:** $\quad 2gL < v_0 < \quad 5gL$ **(Intermediate energy, detachment occurs)**

Since $v_0^2 > 2gL$, the particle crosses the horizontal position $\theta = \pi/2$ with positive velocity.

In the upper semicircle ($\pi/2 < \theta < \pi$), $\cos\theta < 0$.

Evaluate the detachment angle $\cos\theta_d$:

$$\cos\theta_d = \frac{2gL - v_0^2}{3gL}$$

Because $2gL < v_0^2 < 5gL$:

$$-1 = \frac{2gL - 5gL}{3gL} < \cos\theta_d < \frac{2gL - 2gL}{3gL} = 0$$

So $\cos\theta_d \in (-1, 0)$, meaning $\theta_d \in (\pi/2, \pi)$ is physically accessible!

At this point, the speed is:

$$v^2(\theta_d) = v_0^2 - 2gL\left(1 - \frac{2gL - v_0^2}{3gL}\right) = v_0^2 - 2gL + \frac{2}{3}(2gL - v_0^2) = \frac{v_0^2 - 2gL}{3} > 0$$

Conclusion: The detachment point **is reached**. The particle leaves the circular track at angle $\theta_d = \arccos\left(\frac{2gL - v_0^2}{3gL}\right)$ and transitions into free parabolic flight under gravity alone.

- **Regime 3:** $v_0 \geq \quad 5gL$ **(High energy, complete vertical loops)**

At the very top of the circle ($\theta = \pi, \cos\pi = -1$):

$$N(\pi) = \frac{mv_0^2}{L} - mg(2 - 3(-1)) = \frac{mv_0^2}{L} - 5mg \geq 0$$

The reaction force remains positive throughout the entire loop.

Conclusion: The particle **never leaves** the circumference and performs complete revolutions.

Part (b): Rotating Circumference with Angular Velocity ω

1. Equations of Motion in the Rotating Frame

> **Pedagogical Justification:** When the circular track rotates at constant angular speed ω about the vertical diameter Oz, we analyze the motion in the rotating frame $\mathcal{F}_1$. The centrifugal force $\mathbf{F}_{cf} = m\omega^2\boldsymbol{\rho}$ adds a horizontal outward component.

Let cylindrical coordinates be (ρ, ϕ, z), with $\rho = R\sin\theta$, $z = -R\cos\theta$.

In the rotating frame, the unit tangent along increasing θ is:

$$\mathbf{e}_\theta = \cos\theta\,\mathbf{e}_\rho + \sin\theta\,\mathbf{k}$$

Forces acting on the particle:

1. Gravity: $\mathbf{F}_g = -mg\mathbf{k}$.

2. Centrifugal force: $\mathbf{F}_{cf} = m\omega^2\rho\,\mathbf{e}_\rho = m\omega^2 R\sin\theta\,\mathbf{e}_\rho$.

3. Reaction force $\mathbf{N}$ (normal to the wire, along $\mathbf{e}_r$ and $\mathbf{e}_\phi$).

Projecting along the tangent $\mathbf{e}_\theta$:

$$mR\ddot{\theta} = (\mathbf{F}_g + \mathbf{F}_{cf}) \cdot \mathbf{e}_\theta$$

$$\mathbf{F}_g \cdot \mathbf{e}_\theta = (-mg\mathbf{k}) \cdot (\cos\theta\,\mathbf{e}_\rho + \sin\theta\,\mathbf{k}) = -mg\sin\theta$$

$$\mathbf{F}_{cf} \cdot \mathbf{e}_\theta = (m\omega^2 R\sin\theta\,\mathbf{e}_\rho) \cdot (\cos\theta\,\mathbf{e}_\rho + \sin\theta\,\mathbf{k}) = m\omega^2 R\sin\theta\cos\theta$$

Therefore:

$$mR\ddot{\theta} = -mg\sin\theta + m\omega^2 R\sin\theta\cos\theta$$

Divide by mR:

$$\ddot{\theta} + \frac{g}{R}\sin\theta - \omega^2\sin\theta\cos\theta = 0$$

$$\ddot{\theta} + \left(\frac{g}{R} - \omega^2\cos\theta\right)\sin\theta = 0$$

2. Computation of Equilibrium Locations

> **Pedagogical Justification:** Equilibrium configurations in the rotating frame correspond to static solutions where $\ddot{\theta} = 0$ and $\dot{\theta} = 0$.

Setting the angular acceleration to zero:

$$\left(\frac{g}{R} - \omega^2 \cos\theta\right) \sin\theta = 0$$

This equation is satisfied if either factor vanishes:

1. **Vertical Alignments** ($\sin\theta = 0$)**:**

$$\theta_1 = 0 \quad \text{(lowest point / bottom)}$$

$$\theta_2 = \pi \quad \text{(highest point / top)}$$

2. **Off-Axis Equilibrium** ($\cos\theta = \frac{g}{\omega^2 R}$)**:**

This solution exists physically if and only if:

$$\left|\frac{g}{\omega^2 R}\right| \leq 1 \iff \omega^2 \geq \frac{g}{R}$$

When $\omega > \quad \sqrt{\frac{g}{R}}$, two symmetric equilibrium positions emerge:

$$\theta_{3,4} = \pm \arccos\left(\frac{g}{\omega^2 R}\right)$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Supercritical Pitchfork Bifurcation:**
- For slow rotation $\omega < \omega_c = \quad \sqrt{g/R}$, the bottom $\theta = 0$ is the only stable equilibrium.
- When the rotation rate crosses the critical threshold ω_c, the bottom becomes unstable, and the particle naturally climbs up the wire to settle at the tilted angle $\theta_0 = \arccos\left(\frac{g}{\omega^2 R}\right)$, where the upward component of the centrifugal force balances gravity.

- As $\omega \to \infty$, $\cos\theta_0 \to 0 \implies \theta_0 \to \pi/2$ (the particle moves toward the equator).
- **Dimensional check:** $[N] = [mv_0^2/L] = \mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2} = \mathrm{N}$; $\cos\theta_d = (2gL - v_0^2)/(3gL)$ is dimensionless; $\omega_c = \sqrt{g/R}$ has units s^{-1}.
- **Limits of the regimes:** at $v_0^2 = 2gL$, $\cos\theta_d = 0$ ($\theta_d = \pi/2$), the boundary between oscillation and detachment; at $v_0^2 = 5gL$, $\cos\theta_d = -1$ ($\theta_d = \pi$) and $N(\pi) = 0$, the boundary between detachment and full loops. For $\omega \to 0$ in part (b) the equation reduces to the simple pendulum $\ddot{\theta} + (g/R)\sin\theta = 0$.
- **Numerical example ($g = L = 1$, $v_0^2 = 3.5$, regime 2):** $\cos\theta_d = (2 - 3.5)/3 = -0.5$, $\theta_d = 120^\circ$; $v^2(\theta_d) = (v_0^2 - 2gL)/3 = 0.5$ and $N(\theta_d) = v_0^2/L - g(2 - 3\cos\theta_d) = 3.5 - 3.5 = 0$, as required.

`Complete Step-by-Step Solution`

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